

Wazuh Security Monitoring & Threat Hunting Lab

Hands-on security monitoring, endpoint integration, log forwarding, alert investigation and security configuration assessment in an isolated VirtualBox laboratory.

Platform	Wazuh 4.14.2
Security workstation	Kali Linux — 192.168.56.102
Wazuh manager	192.168.56.103
Test system	Metasploitable — 192.168.56.101
Web application	VulnBank
Container platform	Docker
Virtualization	Oracle VirtualBox
Network	192.168.56.0/24

Documentation principle: Completed work, observed results, troubleshooting attempts and unfinished work are clearly distinguished. No successful compromise or detection is claimed where the evidence does not establish it.

1. Project Overview

This project involved building a small security monitoring lab using Wazuh, Kali Linux and Metasploitable, with VulnBank running in Docker as an intentionally vulnerable web application for subsequent security testing.

The primary objective was to gain practical experience with a security monitoring platform and understand how endpoint activity can be collected, monitored and investigated.

- Wazuh manager and dashboard setup
- Wazuh agent configuration
- Network connectivity testing
- Troubleshooting Wazuh agent communication
- Metasploitable log-forwarding configuration
- Security-event monitoring
- Wazuh threat-hunting report analysis
- Security Configuration Assessment (SCA) findings
- CIS benchmark findings
- Basic investigation of Docker/VulnBank logging
- Initial security testing of an intentionally vulnerable system

2. Lab Architecture

The lab used a 192.168.56.0/24 VirtualBox network.

System	Role	IP Address
Metasploitable	Vulnerable test VM	192.168.56.101
Kali Linux	Security workstation / Wazuh agent	192.168.56.102
Wazuh Server	Wazuh manager / dashboard	192.168.56.103

3. Technologies and Tools

Tool	Purpose
Wazuh	Security monitoring, alerting and threat-hunting
Kali Linux	Security testing workstation and Wazuh agent
Metasploitable	Intentionally vulnerable test environment
VulnBank	Intentionally vulnerable web application
Docker	Containerized VulnBank environment
Oracle VirtualBox	Virtual laboratory infrastructure
Linux CLI	System administration and troubleshooting
Syslog	Log forwarding from Metasploitable

4. Wazuh Environment

The Wazuh environment consisted of a Wazuh server running the manager and dashboard components.

The Wazuh environment reported version:

```
Wazuh 4.14.2
```

The Wazuh server network interface was configured with:

```
192.168.56.103/24
```

The Wazuh API/service was observed listening on port 55000. The Wazuh dashboard service was also confirmed to be active and running.

5. Wazuh Agent Configuration

Kali Linux was configured as a Wazuh agent. The agent configuration file was `/var/ossec/etc/ossec.conf`.

During setup, the Wazuh agent initially contained an incorrect placeholder for the manager address:

```
<address>MANAGER_IP</address>
```

This prevented the agent from correctly communicating with the Wazuh manager.

The configuration was corrected to:

```
<address>192.168.56.103</address>
```

Wazuh was then restarted and its status was checked.

6. Network Connectivity Validation

Connectivity was tested in both directions using ICMP.

```
Wazuh server: ping -c 4 192.168.56.102
Kali: ping -c 4 192.168.56.103
```

Both tests returned 4 packets transmitted, 4 received and 0% packet loss, confirming network-level communication between the Kali endpoint and Wazuh manager.

7. Wazuh Manager and Dashboard Validation

The Wazuh manager service was checked using:

```
systemctl status wazuh-manager
```

The service was observed running, with Wazuh manager processes active.

The dashboard was checked using:

```
sudo systemctl status wazuh-dashboard
```

The dashboard service reported an active/running state.

8. Metasploitable Integration

Metasploitable was used as an intentionally vulnerable system within the isolated lab. Its network configuration showed 192.168.56.101.

The system was configured to forward syslog messages toward the Wazuh manager using:

```
*.* @192.168.56.103
```

The logging daemon was restarted with:

```
sudo /etc/init.d/syslogd restart
```

The restart completed successfully.

Scope note: The evidence demonstrates that syslog forwarding was configured and the logging service was restarted successfully. It does not establish that every security-testing action performed against Metasploitable was detected by

Wazuh.

9. Security Testing Against Metasploitable

An SSH security-testing attempt was performed from Kali against the intentionally vulnerable Metasploitable VM.

The attempt encountered an SSH cryptographic compatibility error:

```
no match for method mac algo client->server
```

The connection therefore did not proceed as a successful authentication attempt. This was treated as a troubleshooting observation rather than a successful compromise.

10. Wazuh Threat Hunting

A Wazuh threat-hunting report was generated for 2026-09-07T00:58:55 to 2026-09-08T00:58:55. The report identified the manager as **wazuh-server**.

The report included alert-level evolution, agent activity, MITRE ATT&CK; views, alert summaries, Security Configuration Assessment findings and CIS benchmark findings.

11. Alert Summary

Rule ID	Event	Level	Count
5502	PAM login session closed	3	11
80730	Auditd SELinux permission check	3	9
5501	PAM login session opened	3	7
5402	Successful sudo to ROOT executed	3	5
533	Listened-port status changed	7	4
502	Wazuh server started	3	3
19004	CIS benchmark score below 50%	7	2
5403	First-time user executed sudo	4	2

These results demonstrate that Wazuh was generating alerts for authentication, privilege-related, system and security-configuration events.

12. Security Configuration Assessment

The Wazuh report contained Security Configuration Assessment findings based on CIS benchmarks.

One report entry showed:

```
CIS Distribution Independent Linux Benchmark v2.0.0.  
Score less than 50% (45)
```

This was recorded with level 7 and occurred twice during the reporting period.

The report also identified configuration checks involving areas such as:

- USB storage
- /etc/hosts.deny
- AIDE
- Avahi

- DCCP
- ICMP redirects
- IP forwarding
- IPv6 firewall configuration
- Reverse path filtering
- SCTP
- SELinux
- SSH configuration

13. Additional CIS Findings

The report contained further CIS configuration checks covering areas including automounting, /tmp, CUPS, DHCP, DNS, FTP, HTTP services, IMAP/POP3, LDAP, NFS/RPC, NIS, SELinux/AppArmor, SNMP and SSH configuration.

14. Additional Observed Events

```
501 New wazuh agent connected.  
81102 USB device disconnected
```

The report also recorded an Amazon Linux 2023 CIS benchmark score below 50%, with a score of 44.

15. VulnBank and Docker Environment

VulnBank was deployed as a containerized intentionally vulnerable web application. The Docker environment included a VulnBank web container, PostgreSQL database container and autoheal container.

The VulnBank web application was exposed locally on port 5000. The web container was observed responding to health checks such as GET /healthz with successful responses.

16. Docker Log Investigation

An investigation was performed to determine where the VulnBank container's logs were being stored. The Docker container configuration was inspected rather than assuming a traditional Linux log file.

The Docker log path was identified as a JSON log file under:

```
/var/lib/docker/containers/
```

The log file was verified to exist on the system.

Current status: The Docker log source was successfully identified, but the Docker log was not completed as a Wazuh monitoring source during this phase.

17. Troubleshooting Performed

Issue 1 — Incorrect Wazuh manager address

The agent contained `MANAGER_IP` instead of the actual Wazuh manager address. The address was changed to 192.168.56.103 and Wazuh services were restarted and verified.

Issue 2 — Network connectivity

Connectivity between Kali and the Wazuh manager was tested in both directions and returned 0% packet loss.

Issue 3 — Metasploitable syslog configuration

The initial investigation showed that the expected `/etc/rsyslog.conf` file was not present on the Metasploitable system. The actual `/etc/syslog.conf` file was subsequently identified and used for forwarding.

Issue 4 — SSH compatibility

An SSH security-testing attempt encountered a MAC-algorithm compatibility error. It was recorded as an environmental limitation requiring further troubleshooting.

18. Key Skills Demonstrated

- **Security monitoring:** Wazuh manager/agent architecture, alert monitoring, threat-hunting reports, security event analysis and agent connectivity.
- **Linux:** Command-line administration, network interface inspection, service management, configuration-file modification, log investigation and troubleshooting.
- **Networking:** IPv4 addressing, private lab networks, ICMP connectivity testing and service/port verification.
- **Security operations:** Alert review, authentication-event monitoring, privilege-related event monitoring, SCA and CIS benchmark analysis.
- **Virtualization and containers:** Oracle VirtualBox, Kali, Metasploitable, Docker and container log investigation.

19. What I Learned

Security monitoring is not simply about installing a tool and waiting for alerts. A functional monitoring environment requires correct network configuration, correct agent-manager configuration, service validation, log-source identification, understanding of collected telemetry, contextual alert investigation and careful interpretation of findings.

The troubleshooting process was valuable because configuration errors, missing files and protocol compatibility issues are realistic parts of security operations.

The project also reinforced the importance of distinguishing a confirmed detection from ordinary system activity or an unsuccessful testing attempt.

20. Limitations

- The environment was hosted in VirtualBox.
- The systems were isolated laboratory systems.
- Metasploitable syslog forwarding was configured, but the available evidence does not prove that all generated activity was successfully correlated in Wazuh.
- The SSH testing attempt encountered a compatibility error.
- VulnBank's Docker logs were identified but were not fully integrated into Wazuh during this phase.
- Wazuh threat-hunting data contains many system and configuration events; these should not automatically be interpreted as confirmed attacks.

21. Future Work

- Map the VulnBank application surface.
- Perform controlled vulnerability testing.
- Review application behaviour.
- Use Burp Suite for web-security testing.
- Document confirmed vulnerabilities.
- Investigate whether relevant activity is visible to Wazuh.
- Correlate security-testing activity with available logs.
- Improve Docker log integration where appropriate.

The goal is to connect offensive security testing with defensive monitoring, creating a more complete SOC-style laboratory workflow.

22. Project Outcome

The project established a Wazuh monitoring environment and demonstrated the basic workflow of connecting systems, validating communication, reviewing security telemetry and investigating configuration findings.

- Configured Wazuh manager
- Functioning Wazuh dashboard
- Configured Kali Wazuh agent
- Successful Kali ↔ Wazuh connectivity
- Connected lab environment
- Metasploitable syslog-forwarding configuration
- Wazuh alert and threat-hunting data
- CIS/SCA security findings
- Practical Linux and networking troubleshooting experience
- Initial Docker/VulnBank logging investigation

23. Evidence

Project evidence consists of screenshots and the Wazuh threat-hunting report covering configured systems, network interfaces, connectivity tests, service status, syslog configuration, testing output and Wazuh findings.

The Wazuh report records the monitoring period, manager identity, alert-level information and security findings.

24. Ethical Considerations

All security testing in this project was performed against intentionally vulnerable systems within a controlled virtual laboratory environment.

The purpose was educational: to understand vulnerability assessment, security monitoring, logging and defensive analysis. No authorization was assumed or required for systems outside the laboratory environment.

25. Conclusion

This Wazuh lab provided practical experience with the foundational workflow of a security monitoring environment:

```
Configure
↓
Connect
↓
Validate
↓
Collect
↓
Monitor
↓
Investigate
↓
Document
```

Rather than focusing solely on successful commands, the project emphasized understanding the relationship between systems, network communication, logs, alerts and security findings.

The environment provides a foundation for the next phase: conducting controlled security testing against VulnBank and examining how offensive activity can be investigated from a defensive monitoring perspective.

Portfolio Summary

Project: Wazuh Security Monitoring & Threat Hunting Lab

Primary Platform: Wazuh 4.14.2

Environment: Oracle VirtualBox

Security Workstation: Kali Linux

Monitoring Target: Metasploitable

Web Application: VulnBank

Container Platform: Docker

Network: 192.168.56.0/24

Core competencies: Wazuh · SOC Monitoring · Threat Hunting · Linux · Networking · Syslog · SCA · CIS Benchmarks
· Docker · VirtualBox · Security Troubleshooting